MODULE 02

Python Foundations

Start coding with one language that opens many technology careers

LEARN A SKILL • BUILD PROOF • CREATE YOUR NEXT OPPORTUNITY

Python gives beginners a friendly entry into programming while opening paths toward automation, backend development, data, AI and cybersecurity.

WHO CAN JOIN?

Complete beginners, freshers, non-IT graduates, career switchers and learners who want a versatile first programming language.

WHY STUDENTS CHOOSE THIS PATH

- Beginner-friendly learning path — start with the fundamentals and progress step by step.
- Practical, project-focused training — learn by building instead of only watching lectures.
- Career-oriented skills — connect every topic to real applications and workplace use.
- Portfolio advantage — finish with projects you can discuss in interviews.
- A new direction is possible — your previous degree does not have to decide your entire career.

MODULE CONTENT

Python setup & syntax	Variables & data types
Conditions & loops	Functions
Strings	Lists, tuples, sets, dictionaries
Comprehensions	Lambda/map/filter/reduce
Iterators & generators	Decorators
Exceptions	Modules & packages
Virtual environments	Files
JSON/CSV	Regex
OOP	APIs
Web scraping	Automation
NumPy	Pandas
Matplotlib	Requests
BeautifulSoup	OpenPyXL

HANDS-ON PROJECTS

- Expense Tracker
- Web Scraper
- Automation Toolkit
- Weather/API App
- Data Analysis Project

CAREER DIRECTIONS

Python Developer • Automation Developer • Junior Backend Developer • Data/AI Beginner

YOUR JOURNEY

Learn → Practice → Build → Create a portfolio → Prepare for interviews → Apply your skills in real-world situations.

Note: Training improves skills and employability; employment depends on individual preparation, portfolio quality, interviews and available opportunities.